Name: ____________________________ Class: ____________________________

JURONG JUNIOR COLLEGE
J2 Preliminary Examination 2013

MATHEMATICS **8864**
Higher 1 3 September 2013

3 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 95.
At the end of the examination, fasten all your work together securely, with the cover page in front.

This document consists of **7** printed pages and **1** blank page.

Section A: Pure Mathematics [35 marks]

1 (a) Find the range of values of k for which $kx^2 - 2x + 3k$ is always negative for all real values of x. [3]

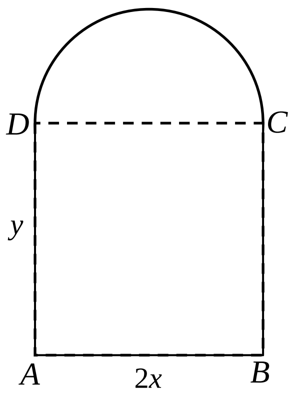
(b) Given that $e^{3x} - 1 = \frac{2}{e^{3x}}$, find the exact value of x. [4]

2 Solve $-2x^3 - 3x^2 > 3 - 8x$. Hence solve $2(\ln x)^3 + 3(\ln x)^2 - 8\ln x + 3 < 0$, giving your answers in exact form. [4]

3 A flower-bed has the shape of a semi-circle joined to a rectangle as shown in the diagram. It is given that $AB = 2x$ m and $AD = y$ m. The flower-bed has a fixed area of 10 m^2. The cost of fencing the straight parts is \$15 per metre and the cost of fencing the semi-circular part is \$20 per metre. Show that the cost, \$$P$, of fencing the perimeter of the flower-bed is given by

$$P = \frac{150}{x} + \left(\frac{60 + 25\pi}{2}\right)x .$$

Using differentiation, find the minimum cost of fencing the flower-bed, giving your answer to the nearest dollar.



[7]

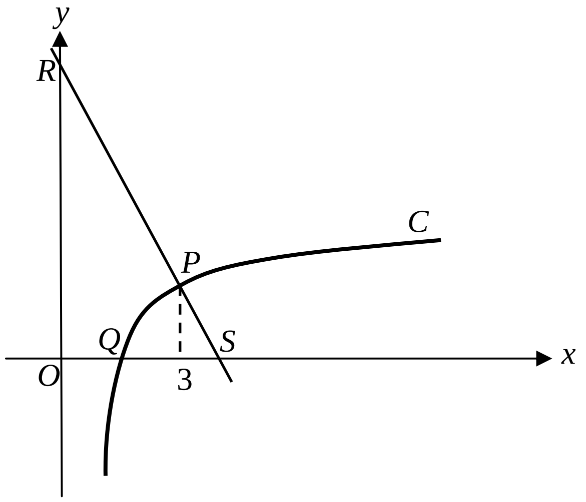
4 (a) Given that $\mathrm{f}(x)=\dfrac{1}{\sqrt{(1-2x)^3}}$.

Find $\int \mathrm{f}(x)\,\mathrm{d}x$ and hence obtain the value of c for which $\int_0^c \mathrm{f}(x)\,\mathrm{d}x = 1.5$. [4]

(b) **(i)** On a single diagram, sketch the graphs of $y=2^{0.5x}+1$ and $y=5-(x-3)^2$, stating clearly the coordinates of the points of intersection. [2]

(ii) Write down as an integral an expression for the area of the region enclosed by $y=2^{0.5x}+1$ and $y=5-(x-3)^2$. Evaluate this integral. [2]

5



The diagram shows the part of the curve C with equation

$$y=4-\frac{a}{2x-1},\ x>\frac{1}{2}.$$

The point P on C has x-coordinate 3. The gradient of the curve C at P is $\dfrac{2}{5}$.

(i) Show that the value of a is 5. [2]

(ii) Find the equation of the normal to the curve C at P. [2]

The curve C meets the x-axis at Q. The normal to C at P meets the x-axis at S and the y-axis at R.

(iii) Find the coordinates of Q. [1]

(iv) Find the exact area of bounded by the curve C, the normal to C at P and the x and y-axes. [4]

Section B: Statistics [60 marks]

6 In a factory, 10% of all the electrical components produced are checked for their quality.

(i) Describe how a systematic sample can be obtained from all the components produced. [2]

(ii) Comment on why this method of sampling gives a better sample compared to choosing the first batch of n components. [1]

7 A game at a fun fair uses a box containing 24 red balls and 1 blue ball. To play the game once, a player takes balls at random from the box, one at a time, without replacement. If the blue ball is taken, the player wins a prize and the game ends. If not, the game ends when 3 red balls have been taken.

Samuel plays the game once.

(i) Draw a tree diagram to represent the possible outcomes. [1]

(ii) Find the probability that Samuel will win a prize. [2]

(iii) Given that Samuel has won a prize, find the probability that he took more than 1 ball. [3]

Three players who each play the game once are chosen at random. Find the probability that only one of them wins a prize. [2]

8 Events A and B are such that $\mathrm{P}(A) = 2p$, $\mathrm{P}(B) = p$ and $\mathrm{P}(A' \cap B') = \frac{8}{15}p$.

(i) Given that $\mathrm{P}(A \mid B) = \frac{2}{3}p$, find a quadratic equation satisfied by p. [4]

(ii) Hence, find p. With this value of p, find the probability that exactly one of A and B occurs. [3]

9 The probability that a seed of a particular variety of bean will germinate when sown is 0.65. Seeds are sold in packets of 12. A packet of seed is selected at random. Calculate the probability that the number of seeds which will germinate when sown is

(i) at least 7, [2]

(ii) between 4 and 9 inclusive. [2]

Twenty-five packets of seeds are selected at random. Find the probability that there are at most 3 packets with fewer than 7 seeds that will germinate. [2]

Ten packets of seeds are selected at random. Using a suitable approximation, estimate the probability that more than 90 seeds will germinate. [4]

10 A company manufactures jam and puts it into jars. The mass of jam in a jar is normally distributed with mean mass m and standard deviation 6.5 grams. The jars are labelled as containing at least 320 grams of jam.

(i) If the company wishes to have not more than 1 jar in 1000 to contain less than 320 grams of jam, show that the least value of m for this to be achieved, to 2 decimal places, is 340.09. [2]

(ii) The mass of an empty jar (with lids) is normally distributed with mean 160 grams and standard deviation 4 grams. Assuming that the masses of the jars and the jam are independent, find the probability that a randomly chosen jar of jam will have a mass of at least 505 grams. [2]

(iii) A sample of 15 jars of jam is randomly chosen. Find the probability that the mean mass of the sample is more than 498 grams. [3]

(iv) Jars of jam from this company are packed in large cartons of 12 and small cartons of 6. The mass of an empty large carton is normally distributed with mean 50 grams and standard deviation 1 gram while the mass of an empty small carton is normally distributed with mean 20 grams and standard deviation 1 gram. Assuming the masses of jars, jam and small/large cartons are independent, find the probability that the mass of a randomly chosen large carton of jam is within 10 grams of twice the mass of a randomly chosen small carton of jam. [5]

11 A company that produces a brand of beef burgers claimed that its burgers have an average percentage fat content of not more than 12 percent. A consumer association wished to investigate if the claim is true and measured the percentage fat content (X) of 50 randomly chosen burgers. The results are summarised as follows:

$$\sum(x-12)=105, \quad \sum(x-12)^2=2250.$$

(i) Find unbiased estimates of the population mean and variance. [3]

(ii) Test at the 2% level of significance whether there is sufficient evidence to refute the company's claim? [4]

(iii) Explain what is meant by the phrase "2% level of significance" in this context. [1]

(iv) State, with a reason, whether, in using the above test, it is necessary to assume that the percentage fat content of the beef burgers has a normal distribution. [1]

Measurements taken on another random sample of 50 burgers give an average percentage fat content of F percent. A test at the 5% level of significance indicates that the company's claim is untrue. Assume that the population standard deviation of the percentage fat content is 6.5. Obtain an equation for the least possible value of F, and solve it. [3]

[Turn over

12 The following table shows how the time t, in minutes taken for a chemical reaction varies with the amount of catalyst x grams added to a reaction mixture with the same amount of reactants.

x (g)	0	0.5	0.8	1.0	1.2	1.5	2.0
t (min)	120	98	90	81	64	59	40

(i) Draw a sketch of the scatter diagram for the data, as shown on your calculator. [2]

(ii) Calculate the product moment correlation coefficient and comment on its value in the context of the data. [2]

(iii) Find the equation of the regression line of t on x, giving your answer to 4 decimal places. Sketch this line on the scatter diagram. [2]

(iv) Estimate the amount of catalyst used if the time taken for the chemical reaction to complete is 1 hour, leaving your answer to 2 decimal places. Comment on the reliability of the estimate. [2]

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